Session 5: Solutions and facilitation notes

Foundations of Clinical Research · instructor copy (do not distribute before the session)

Exercise numbers follow `Practicals/s5_exercises.md`. All FLUID-ICU numbers come from `Course/Data/key_results.md` (simulated data).

Exercise 1: Classify the variables

#	Variable	Answer	Explanation
a	Age (years)	**Continuous**	Measured on a scale. Keep the exact number on the CRF (or collect date of birth).
b	Sex	**Binary**	Two categories. Coding 1/2 does not make it numerical.
c	Glasgow Coma Scale	**Ordinal**	Ordered scores (3–15) with unequal steps. Often *analysed* like a number, which is acceptable if stated, but it is ordinal by nature.
d	Source of sepsis	**Nominal**	Several categories, no natural order.
e	Number of vasopressors	**Count**	Whole numbers 0–3. Accept "ordinal" if the participant argues it well; not continuous.
f	Cumulative fluid balance (ml/kg)	**Continuous**	Can be negative or positive, with any value on the scale. The Paper 1 primary outcome.
g	Alive at day 28	**Binary**	Yes/no.
h	Time to death, censored at day 28	**Time-to-event**	Two pieces of information: how long, and whether the event happened. Analysed with Kaplan–Meier / Cox.
i	Pain none–severe	**Ordinal**	Ordered categories.
j	Blood group	**Nominal**	No order.

Bonus: any continuous variable (age, fluid balance, lactate). You can always group later, but you cannot recover the exact value from a group.

Common errors: GCS called continuous; time-to-death called continuous (ignores censoring); number of vasopressors called continuous.

Exercise 2: Fix the bad CRF items (model answers)

#	Bad item	Problem	Model rewrite
a	Fluid given: ____	No unit, no time window, no definition (crystalloid? blood? drug diluents?)	"Total IV fluid received from ICU admission to 24 h (ml), including crystalloids, colloids and drug diluents: [ ][ ][ ][ ][ ] ml □ not recorded"
b	Weight: ____	No unit; measured or estimated? when?	"Body weight at ICU admission (kg): [ ][ ][ ].[ ] □ measured □ estimated □ not available"
c	Comorbidities: _____	Free text: impossible to analyse, spelling varies	Tick list: "Comorbidities present at admission (tick ALL that apply): □ diabetes □ chronic kidney disease □ heart failure □ COPD □ cirrhosis □ cancer □ none □ unknown"
d	Lactate high? Y / N	Opinion ("high" by whose cut-off?), loses the value, no time point	"Serum lactate at ICU admission (mmol/L): [ ][ ].[ ] □ not done" (define "high" later, in the analysis)

Accept any version that passes the four checks (unit, time point, one reading, unknown option). Praise answers that notice that collecting the *number* is better than collecting a yes/no.

Exercise 3: Choose the right summary

#	Variable	Shape / type	Summary	Graph
a	Age, bell-shaped	Symmetric continuous	**Mean (SD)**	Histogram
b	ICU length of stay	Right-skewed	**Median (IQR)**	Histogram or boxplot
c	Died in hospital	Binary	**n (%)**	Bar chart
d	CRP	Right-skewed	**Median (IQR)**	Histogram / boxplot
e	Source of sepsis	Nominal	**n (%) per category**	Bar chart (not 3-D pie)
f	SBP by treatment group	Symmetric, two groups	**Mean (SD) per group**	Side-by-side boxplots

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Exercise 4: Judge a bad figure

1. Each bar is the **mean** fluid balance per group. The error bar is the **standard error (SE)**, shown in one direction only. Missing: the individual patients, the spread (SD or IQR), the shape (skew), the n per group on the figure, and what the error bar represents.

2. Any two of: the data are **skewed**; **one extreme patient** (160 ml/kg) pulls the usual-care mean up; the groups **overlap** a lot; the median; the number of patients; negative values.

3. Dots (one per patient) plus a boxplot per group (see the "dots + box" slide), y-axis labelled with units, n under each group. You gain: shape, spread, outliers and overlap become visible, and the reader can judge the comparison.

4. The truncated axis makes a 2-percentage-point difference (31% vs 29%) look like a three-fold difference. Fix: start bars at 0 and label the values (or show the difference with its 95% CI).

5. (i) The author chose the two scales, so the lines can be made to cross or move together wherever wanted; (ii) two trends over time moving together is **correlation, not causation** (other changes may have happened at the same time). Better: two panels sharing the time axis, and an appropriate design (e.g. interrupted time series, Session 3).

Exercise 5: [FLUID-ICU] Spot the problems in the raw FLUID-ICU data

(See `Course/Data/cleaning_log.md` for the full log.)

#	What is wrong	What the script does
a	One variable coded many ways (and a leading space)	Recode to `Female` / `Male`
b	Dates in different formats (03/04: March or April?)	Read each format explicitly and convert to one date format (YYYY-MM-DD)
c	ICU C reported lactate in mg/dL, not mmol/L (about 9 times larger; found by comparing medians by ICU)	Divide ICU C values by 9.01 (1 mmol/L = 9.01 mg/dL); confirm with the site laboratory
d	`999` is a missing-value code, not a lactate	Set to missing (NA)
e	Impossible age (outside the allowed range)	Set to missing, send a query to the site; keep the patient
f	Duplicate record	Check the two rows are identical, keep one copy (623 rows become 620 patients)

Bonus: editing by hand is invisible and not reproducible. A script plus a log lets anyone check and repeat every change. The log records the problem, how many records, what was done, and why (plus queries sent).

Exercise 6: [FLUID-ICU] Read the FLUID-ICU Table 1

All numbers from `Course/Data/key_results.md` (simulated data).

1. Negative balance n = 245, positive n = 373, overall 618. Two of the 620 enrolled patients have no computable day-3 fluid balance (an impossible weight or fluid output was set to missing during cleaning), so they cannot be placed in a group.

2. Expected answers: mean (SD) for age, which is roughly symmetric (64.6 vs 61.4 years); median (IQR) for SOFA, a skewed score (6 (4 to 8) vs 8 (6 to 10)); median (IQR) for lactate, which is right-skewed (2.2 (1.4 to 3.0) vs 2.9 (1.9 to 4.0) mmol/L); n (%) for vasopressor, ventilation, heart failure and sex.

3. Severity: SOFA 8 vs 6; vasopressor 205/373 (55.0%) vs 61/245 (24.9%); mechanical ventilation 216/373 (57.9%) vs 97/245 (39.6%). (Heart failure goes the other way: 43/373 (11.5%) vs 54/245 (22.0%), because clinicians give less fluid in heart failure.)

4. Lactate: 91 missing among the 618 (93 of all 620 enrolled, 15.0%).

5. Model sentences: "Of 620 enrolled patients, 618 had a computable day-3 fluid balance, of whom 373 (60.4%) had a positive balance. Patients with a positive balance were more severely ill at admission (median SOFA 8 vs 6; vasopressor 55.0% vs 24.9%; mechanical ventilation 57.9% vs 39.6%)."

6. Larger: the positive-balance group was sicker, and sicker patients die more, so the crude association mixes fluid with severity (confounding by indication). Confirmed in Session 6: crude OR 2.49 vs adjusted 1.62.

Exercise 7: [FLUID-ICU] A FLUID-ICU question as a PECO sentence

In class, everyone fills in the table for Q1 alone, then a few people read their sentence. Participants fill it in again after choosing their own question at the end of Session 6.

The table for Q1

	Model answer
The question (Q1 to Q5), copied from page 1 of its results pack	Q1: In adults with sepsis in the ICU, is a positive cumulative fluid balance by day 3 associated with higher 28-day mortality?
P: population	Adults with sepsis admitted to one of the four ICUs
E: exposure (variable name in the data dictionary) and its type	Positive fluid balance by day 3, `positive_balance_d3` (binary). It is derived from `cum_fluid_balance_d3_ml_kg` (continuous)
C: comparator	A negative or zero fluid balance by day 3
O: outcome (variable name) and its type	Death from any cause by day 28, `died_28d` (binary). Also time to death, `time_to_event_28d` (time-to-event)
One-sentence research question	"In adults with sepsis admitted to the ICU, is a positive cumulative fluid balance by day 3, compared with a negative balance, associated with higher 28-day mortality?"
Which test will compare the groups? Which model will adjust?	Chi-square test for the crude comparison. Logistic regression to adjust. Kaplan-Meier and Cox regression for time to death

When you redo the table for your own question (Q2 to Q5)

Q	E: exposure (type)	O: outcome (type)	Test, then model
Q2	Cumulative balance, `cum_fluid_balance_d3_ml_kg` (continuous, per 10 ml/kg)	AKI, KDIGO stage 1 or more, `aki_any` (binary)	Chi-square by group, then logistic regression

Q	E: exposure (type)	O: outcome (type)	Test, then model
Q3	Cumulative balance (continuous)	Ventilator-free days to day 28, `vent_free_days_28` (count 0 to 28, skewed, piled at 0 and 28)	Mann-Whitney, then linear regression
Q4	Lactate at admission, `lactate_mmol_l`, mmol/L (continuous, 15% missing)	Death by day 28, `died_28d` (binary)	Logistic regression, complete case and multiple imputation
Q5	Positive balance, `positive_balance_d3` (binary)	ICU length of stay in survivors, `icu_los_days` (continuous, right-skewed)	Mann-Whitney, then linear regression on the log scale

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Exercise 8: Proposal builder 5 (what a good answer looks like)

Variables table

Role	Variable	How measured	Type
Primary outcome	Death by day 28	Hospital records / phone call at day 28	Binary
Secondary outcome	Cumulative fluid balance at day 5	Fluid charts, ml/kg of admission weight	Continuous
Intervention	Restrictive vs usual fluid strategy	Randomised allocation	Binary
Baseline / confounders	Age, SOFA score, lactate at admission, source of sepsis	CRF at ICU admission	Continuous, ordinal-like score, continuous, nominal

Sample size: 40% vs 30% mortality, alpha 5% two-sided, power 80% gives 356 per group (spreadsheet sheet 2). Dividing by (1 − 0.10) gives 396 per group, 792 in total.

Analysis plan: describe with mean (SD) / median (IQR) / n (%); compare 28-day mortality with a chi-square test; report the risk ratio and risk difference with 95% CIs; fluid balance compared with Mann–Whitney and reported as median (IQR) per group. (Observational designs: add "adjusted for age, SOFA and lactate using logistic regression".)

Quick check for the instructor (while modelling, and when reading drafts sent by email):

Element	Good enough	Needs work
Variables table	Outcome, exposure and ≥ 3 confounders, all with a type	Confounders that are actually outcomes; no types
Mini-CRF	6–10 items, units, time points, unknown option	> 15 items; free text; no units
Data dictionary	3–5 rows with codes and allowed ranges	Names with spaces, no codes
Sample size	4 inputs with a stated source	"All patients in one year"
Analysis plan	Summary + test + effect measure with 95% CI	"Data will be analysed with SPSS"

Data management (good enough): EDC with range and logic checks at entry (or paper with double entry of key variables); coded study IDs with the linking log kept separately and locked; audit trail/cleaning log; raw export read-only, dated versions, encrypted 3-2-1 backups; data-sharing statement in the consent form.